Effects of Forces — Turning, Pressure, Centre of Mass

📂 1.6 Effects of Forces 🎯 IGCSE Physics 0625 — Extended 📝 Version 1 📅 2026-06-30

📜 📜 History

The study of **turning effects** (moments) began with **Archimedes**, who famously said: *"Give me a lever long enough and a place to stand, and I shall move the Earth."*

Levers have been used since prehistoric times, but Archimedes was the first to mathematically describe how they work — the ratio of the lengths of the lever arms determines the mechanical advantage.

Pressure was studied by **Blaise Pascal (1623–1662)**, a French mathematician and physicist. Pascal's Principle states that pressure applied to an enclosed fluid is transmitted equally in all directions — the foundation of hydraulic systems.

Centre of mass was formalised by **Archimedes** and later expanded by Newton to describe how objects balance and rotate.

Did you know? The Leaning Tower of Pisa leans because the centre of mass of the tower has shifted past its base of support. Engineers have been working for centuries to prevent it from toppling! The current angle is about 4°, and recent stabilisation work should keep it standing for another 200 years.

🌍 🌍 Nature & Applications

- 🔧 **Levers everywhere**: Scissors, pliers, bottle openers, seesaws, crowbars
- 🚲 **Bicycle brakes**: A small force on the brake lever creates a much larger force on the brake pads
- 🔪 **Sharp knives**: Smaller area = larger pressure = easier cutting
- ❄️ **Snowshoes**: Large area = small pressure = don't sink into snow
- 🚗 **Hydraulic brakes**: Pascal's Principle amplifies the driver's foot force
- 🪂 **Centre of mass**: Skydivers adjust their body position to control rotation
- 🏗️ **Stability**: Wide-base objects (e.g., a tall lamp with a heavy base) are more stable

⚛️ ⚛️ Theory

Turning Effect (Moment of a Force)

The turning effect of a force is called a **moment** (or torque).

Where:
- = moment (Nm)
- = force (N)
- = perpendicular distance from the pivot to the line of action of the force (m)

Principle of Moments: For an object in equilibrium (balanced):

Levers

A lever is a rigid bar that rotates around a **pivot** (fulcrum). There are three classes:

Class	Arrangement	Example
1st	Fulcrum between effort and load	Seesaw, crowbar
2nd	Load between fulcrum and effort	Wheelbarrow, nutcracker
3rd	Effort between fulcrum and load	Tweezers, fishing rod

Mechanical advantage:

Pressure

Pressure is the **force per unit area**.

Where:
- = pressure (Pa or N/m²)
- = force (N)
- = area (m²)

Pascal (Pa):

Pressure in Liquids

Where:
- = pressure (Pa)
- = depth (m)
- = density of the liquid (kg/m³)
- = gravitational field strength (N/kg)

Key properties of liquid pressure:
- Increases with **depth**

- Acts **equally in all directions** at a given depth
- Depends on **density** of the liquid
- Does NOT depend on the shape of the container

Centre of Mass

- **Centre of mass** is the point where the entire mass of an object can be considered to act
- For a **regular, uniform** object, the centre of mass is at its **geometric centre**
- For an **irregular** object, find it by **suspension method** (plumb line)

Stability and centre of mass:
- An object is **stable** if its centre of mass lies above its **base of support**
- To topple an object, the centre of mass must move **outside** the base
- **Wide base + low centre of mass** = most stable

$$M = F \times d$$

$$\text{Sum of clockwise moments} = \text{Sum of anticlockwise moments}$$

$$\Sigma M_{\text{clockwise}} = \Sigma M_{\text{anticlockwise}}$$

$$\text{MA} = \frac{\text{load}}{\text{effort}} = \frac{\text{effort distance}}{\text{load distance}}$$

$$P = \frac{F}{A}$$

$$P = h\rho g$$

M F d P F A $1\ \text{Pa} = 1\ \text{N/m}^2$ P h ρ g

✍️ ✍️ Worked Examples

Example 1: Moment of a Force

A force of 20 N is applied perpendicularly to a spanner at a distance of 0.3 m from the nut. What is the moment?

Solution:

Example 2: Principle of Moments

A 2.0 m long seesaw has a pivot at its centre. A child of mass 30 kg sits 0.8 m from the pivot on the left. Where must a child of mass 40 kg sit on the right to balance?

Solution:
Weight of left child: N
Weight of right child: N

Clockwise moment (right child) = Anticlockwise moment (left child)

The 40 kg child must sit **0.6 m** from the pivot.

Example 3: Pressure

A box weighing 600 N rests on the floor. The base of the box measures 0.5 m × 0.4 m. What pressure does it exert?

Solution:

Example 4: Pressure in Water

A submarine is at a depth of 50 m in seawater (kg/m³). What is the water pressure? (N/kg)

Solution:

Total pressure on the hull includes atmospheric pressure (Pa) on top:

$$M = Fd = 20 \times 0.3 = 6.0\ \mathrm{Nm}$$

$$294 \times 0.8 = 392 \times d$$

$$d = \frac{294 \times 0.8}{392} = 0.6 \text{ m}$$

$$A = 0.5 \times 0.4 = 0.2 \text{ m}^2$$

$$P = \frac{F}{A} = \frac{600}{0.2} = 3000 \text{ Pa}$$

$$P = h\rho g = 50 \times 1025 \times 9.8$$

$$P = 502\,250 \text{ Pa} \approx 5.02 \times 10^5 \text{ Pa}$$

$$P_{\text{total}} = 5.02 \times 10^5 + 1.01 \times 10^5 = 6.03 \times 10^5 \text{ Pa}$$

$W_1 = 30 \times 9.8 = 294$ $W_2 = 40 \times 9.8 = 392$ $\rho = 1025$ $g = 9.8$ $\approx 1.01 \times 10^5$